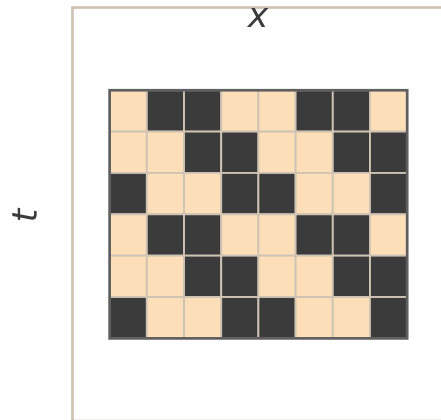


Relative-periodic domain selection by Bernoulli NML

1 Observed space-time diagram

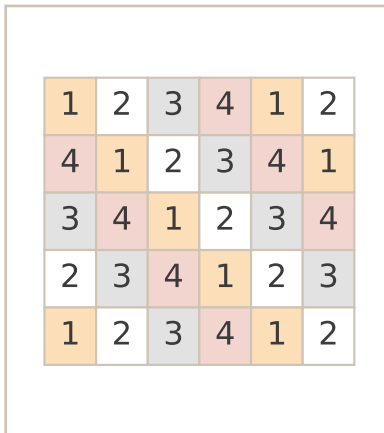
Choose candidate (p, s) .
Example: $p = 2, s = 1$.



Raw CA space-time diagram to explain.

2 Translation orbits

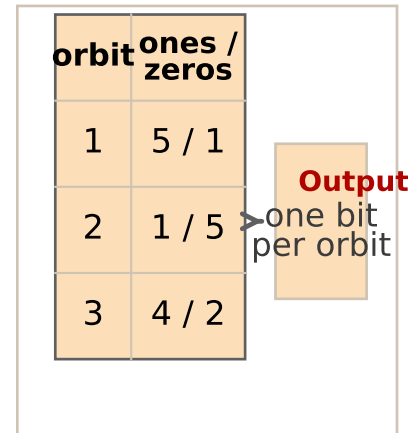
Apply $(t, x) \mapsto (t + p, x + s)$.
Same color = same orbit.



Each orbit contributes one domain-template bit.

3 Majority-vote fit

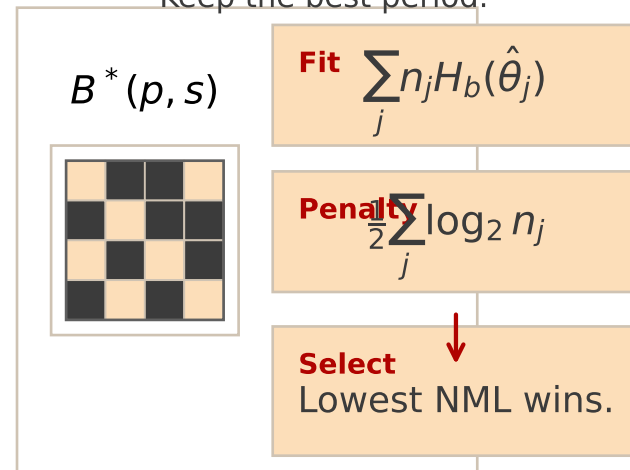
Count ones and zeros in each orbit.
Choose the majority bit.



Defect mask: $M = U \oplus B^*$.

4 Score and select

NML = fit + penalty.
Keep the best period.



Output: periodic domain template + defect mask.